8_23_2026 Update

Boltz-2, Nesso-1, FlashBind, and an FEP+ four-kinase reproduction

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Repository: github.com/Yusheng-cai/BindingAffinityPredictor

Executive summary

This week, I focused on four linked questions:

1. What does Boltz-2 do?
2. How does Nesso-1 change that workflow, and what does it improve or sacrifice?
3. Where does FlashBind's dock-and-score workflow fit relative to these two models?
4. Can I reproduce the reported public affinity benchmark using the authors' released checkpoints?

In brief, Boltz-2 jointly predicts an atom-resolved protein–ligand complex and learned affinity outputs from protein sequence, ligand chemistry (SMILES), and usually an MSA.

Nesso-1 removes the costly atom-coordinate diffusion stage and the MSA requirement. It instead predicts affinity from jointly processed protein–ligand features and a predicted pocket. This makes Nesso much more practical for screening, at the cost of not producing the detailed heavy-atom pose supplied by Boltz-2.

FlashBind makes a third architectural choice. Rather than jointly generating an atom-resolved complex with an expensive diffusion decoder, it first obtains a three-dimensional receptor, uses FABind+ to predict a pocket and dock the ligand, and then scores the resulting local protein–ligand geometry with an E(3)-equivariant graph neural network. It is therefore best understood as a two-stage *dock then score* system for virtual screening. The receptor can be experimental or predicted, but three-dimensional receptor coordinates are required internally [1].

For the empirical test, I reconstructed the public 87-compound, four-kinase benchmark and ran the released models on every compound. This is the *affinity-ranking* evaluation: it compares predicted affinity scores with experimental potency labels and does not evaluate coordinates. Nesso-1 achieved a compound-count-weighted mean within-target Pearson correlation of $r = 0.807$ (compound-bootstrap 95% interval 0.731–0.873), essentially reproducing the preprint value of 0.80 [2]. Boltz-2, using the MSA-1024 protocol described below, achieved $r = 0.627$ (interval 0.484–0.744), close to its paper reference of 0.66 [3]. FlashBind's two-checkpoint value ensemble achieved $r = 0.533$ (interval 0.382–0.669), reproducing its paper value of 0.53 [1]. I obtained 87/87 affinity outputs from all three models.

Separately, I performed a *structure-recovery* evaluation against 16 exact experimental protein–ligand cocrystals. Boltz-2 had median global protein Cα, locally fitted pocket Cα, and pocket-aligned ligand RMSDs of **1.231**, **0.400**, and **0.898 Å**, respectively. The released FlashBind/FABind+ poses had a median ligand RMSD of **0.634 Å**; its receptor RMSDs describe the supplied experimental receptor rather than a FlashBind protein prediction.

Headline affinity result: complete 87-compound benchmark

	Weighted Pearson r	95% interval	Paper reference
Nesso-1	**0.807**	0.731–0.873	0.80
Boltz-2 MSA-1024	**0.627**	0.484–0.744	0.66
FlashBind, released poses	**0.533**	0.382–0.669	0.53

87/87 outputs for all three models • **Paired difference:** $+0.180$ $[+0.058, +0.319]$

Largest target-level separation: p38 ($r = 0.869$ Nesso versus $r = 0.364$ Boltz). The Boltz result uses the documented 1,024-row trunk-MSA subsample; the FlashBind result uses released FABind+ poses.

1 Benchmark

I used the older OpenFF Git commit `da7c337` because it contains the full 87-example benchmark used in both model papers: $16 + 16 + 21 + 34 = 87$ neutral protein–ligand examples [4].

Each target defines one separate medicinal-chemistry series: the protein is held fixed while the ligand is changed. The four targets are cyclin-dependent kinase 2 (CDK2), tyrosine kinase 2 (TYK2), c-Jun N-terminal kinase 1 (JNK1), and p38α mitogen-activated protein kinase. All four are protein kinases, but they are different proteins with different ligand series and different biochemical assays.

Target / assay series	Ligands (N)	Protein construct input	Endpoint	Reference PDB
CDK2 (cyclin-dependent kinase 2)	16	CDK2, 303 aa; cyclin A2, 258 aa	IC50	1H1Q
TYK2 (tyrosine kinase 2)	16	TYK2, 302 aa	Ki	4GIH
JNK1 (c-Jun N-terminal kinase 1)	21	JNK1, 370 aa; JIP1 peptide, 11 aa	IC50	2GMX
p38α (MAPK14)	34	p38 kinase, 372 aa	IC50	3FLY

Table 1: Frozen benchmark composition. One model example is one target construct paired with one ligand, so N is the number of independently scored ligands in that target's series.

The columns have the following precise meanings:

Target protein / assay series. The common protein construct against which a chemically related set of ligands was experimentally tested. I calculated each target's correlation separately rather than mixing compounds from different targets.

Ligands (N). The number of distinct protein–ligand examples in that series. For example, CDK2 has 16 ligands. Nesso therefore received 16 CDK2 inputs: each contained the same CDK2–cyclin A2 sequences but a different ligand SMILES. The four values sum to $16 + 16 + 21 + 34 = 87$ model examples.

Protein construct. The exact amino-acid sequence or sequences given to the model. "303 aa; 258 aa" means two chains, not one 561-residue protein: a 303-residue CDK2 chain and its 258-residue cyclin A2 partner. Similarly, the JNK1 input includes the 11-residue JIP1 peptide found in the curated reference complex.

Experimental endpoint. The laboratory measurement used as the benchmark's ground-truth label. IC50 is the concentration that produces 50% inhibition under a particular assay protocol. Ki is an inhibition constant inferred through an enzyme-kinetic model. Smaller IC50 or Ki means stronger apparent inhibition. These endpoints are related but not interchangeable, and neither should automatically be called a binding free energy. Source values in nM or μM were converted to μM and then to $\log_{10}(\text{value}/\mu\text{M})$ before comparison. CDK2, JNK1, and p38 use IC50 labels; TYK2 uses Ki labels.

Reference PDB. The Protein Data Bank accession for the experimentally solved complex used by the curated benchmark to identify the target and its exact construct. For example, `1H1Q` identifies the reference CDK2–cyclin A2 complex. In this calculation, the PDB coordinates and bound ligand pose were *not* given to Nesso or Boltz; only the deposited protein sequence and the benchmark provenance were taken from this record. FlashBind is different: its released-pose calculation uses the corresponding fixed protein coordinates, but not the crystallographic ligand pose.

2 Model description: Boltz-2, Nesso-1, and FlashBind

2.1 Boltz-2: cofold and score

At the highest level, Boltz-2 is a biomolecular cofolding model with an additional affinity-prediction pathway. One example can contain protein sequence(s), ligand chemistry, an MSA, and optional templates or constraints. Its structure trunk integrates these inputs, and a stochastic diffusion decoder generates three-dimensional coordinates for the modeled heavy atoms. The released system returns sampled complex structures and confidence quantities; for protein–ligand inputs it can also return a binder probability and continuous affinity estimates [3]. These learned affinity outputs are assay-calibrated model scores. They are not direct alchemical calculations of an equilibrium ΔG_{bind}.

2.2 Nesso-1: predict affinity without coordinate diffusion

Nesso-1 follows a different affinity workflow. It does not run the atom-coordinate diffusion decoder used by Boltz-2, and it supplies protein context through pretrained ESM-2 embeddings instead of an MSA.

The Nesso paper reports more than a tenfold inference speed-up over Boltz-2, and more than twentyfold on some larger targets, while matching or exceeding Boltz-2 on its evaluated affinity benchmarks [2]. Nesso does not return the atom-resolved complex produced by Boltz-2, so the two models serve different structural needs. For this project, I compare their released affinity outputs on identical protein and ligand inputs. Neither affinity head is a molecular simulation or a direct calculation of ΔG_{bind}; both are learned from experimental potency and affinity labels.

2.3 FlashBind: dock then score

At the user-facing level, FlashBind can begin with a protein sequence and a ligand SMILES. Internally, however, the sequence must first be resolved to a three-dimensional receptor. The released preprocessing searches for an experimental or predicted structure and can fall back to a structure predictor when one is unavailable. FlashBind then applies the following procedure [1]:

1. **Resolve the receptor.** Use a PDB structure, a predicted structure such as an AlphaFold model, or another generated receptor model. A crystal structure is useful but not mandatory.

2. **Predict the pocket and pose.** FABind+ rapidly identifies a candidate binding region and places the ligand in that region.

3. **Construct the local complex.** The pipeline crops the receptor around the proposed ligand pose and represents the local protein–ligand environment as a geometric graph.

4. **Score the geometry.** An E(3)-equivariant graph neural network processes the local complex. Equivariance ensures that rotating or translating the coordinates does not arbitrarily change the binding prediction.

5. **Read out the prediction.** Separate heads can return a binary activity probability or a continuous assay-derived affinity score. The continuous score is not a thermodynamic calculation of ΔG_{bind}.

This modularity separates pose generation from affinity scoring, which makes large-library screening more practical but also creates a sequential error path:

$$\text{receptor conformation} \longrightarrow \text{pocket selection} \longrightarrow \text{docked pose} \longrightarrow \text{binding score.}$$

An incorrect receptor or docked pose can therefore limit the scorer even if the final neural network is otherwise well calibrated.

Model workflows at a glance

Boltz-2: sequence + ligand + MSA → joint trunk + atom diffusion → atom-resolved complex + affinity

Nesso-1: sequence + ligand → ESM-2 + pair trunk + predicted pocket → affinity, without a final Cartesian complex

FlashBind: 3D receptor + ligand → FABind+ pocket and pose → local EGNN score

3 Reproducing the benchmark

3.1 Nesso-1

I ran Nesso using the official source at Git revision `f0156e9`, released checkpoint tag `v1.0.0` (resolved Hugging Face revision `1896c84`), and ESM-2 `facebook/esm2_t33_650M_UR50D` at revision `08e4846`. I used seed 42, five recycling steps, bfloat16 mixed precision, the default two-stage 22 Å protein-pocket crop with a 256-token budget, and disabled optional CUDA kernels. I generated each unique protein-chain embedding once per target and reused it across the target's ligands. I retained the raw continuous affinity values, the two component estimates, binder probabilities, and entropy diagnostics before analysis.

Nesso does not generate an all-atom protein–ligand complex. It combines sequence-derived protein features and ligand atom/bond features in a geometric trunk, predicts an interaction region, and reads out affinity. This is why it can screen faster than a full coordinate-diffusion model [2].

3.2 Boltz-2 inference protocol

I used Boltz 2.2.1 source revision `b1ebfc4`, the released structure and affinity checkpoints, seed 42, three structure recycles, one 200-step structure sample, and five 200-step affinity samples. I generated one ColabFold/MMseqs2 MSA per unique protein chain and reused it across every ligand for that target. I capped each MSA at the frozen maximum of 8,192 parsed rows. I did not use templates, experimental coordinates, or the optional molecular-weight correction.

Normally, all retained MSA rows—up to that 8,192-row cap—can enter the model trunk. That full-depth calculation exceeded the memory of the local 10 GiB GPU, so I enabled Boltz's documented `--subsample_msa` option and used 1,024 randomly selected MSA rows in the trunk [3]. This reduces memory use while preserving the larger parsed MSA as the sampling pool.

I kept every other setting unchanged and applied this MSA-1024 protocol to all four targets. I obtained 87/87 structures and affinity predictions. This is a released-checkpoint reproduction with reconstructed preprocessing and a documented MSA-depth choice, not the authors' exact evaluation pipeline.

3.3 FlashBind released-pose scoring run

I used FlashBind source revision `f161268`, the two released value checkpoints at Hugging Face revision `fa6362c`, and the released FEP+4 archive at revision `50b1511`. The archive contains exactly 87 IDs, four fixed receptor PDBs, one FABind+-predicted ligand pose per pair, ESM3 protein representations, and TorchDrug ligand representations. I matched all 87 molecules to my canonical manifest by target and InChIKey. Every experimental label matched to floating-point precision, and the four receptor PDB files were byte-identical to my local benchmark copies.

To remain compatible with the local NVIDIA driver, I used an isolated PyTorch 2.5.1/CUDA 12.1 scoring environment rather than the upstream PyTorch 2.7.1/CUDA 12.6 specification. I retained the authors' inference code, checkpoint weights, input features, and numerical settings, but this dependency change means the calculation is not a bitwise environment reproduction.

I first ran one CDK2 ligand through both checkpoints and required finite `pred_value`, `pred_value_raw`, and molecular weight fields. After that check passed, I ran all 87 records on one GPU with the authors' 20 Å local distance threshold and averaged the two checkpoint outputs. I used the calibrated `pred_value`, documented as $\log_{10}(\mathrm{IC50/Ki/Kd\ in\ }\mu M)$, for the primary analysis while retaining the uncalibrated values. This experiment uses released FABind+ poses rather than rerunning FABind+; it therefore tests the released scorer and supplied preprocessing, not independent end-to-end docking reproducibility.

4 Metric policy: affinity and structure

I use two distinct metric families that answer different questions. The **affinity metrics** use all 87 compounds and test agreement between model scores and experimental potency labels. The **structure metrics** use only the 16 pairs with deposited crystal structures and compare Boltz-2 with the released FlashBind/FABind+ poses.

4.1 Affinity-ranking metrics on 87 compounds

I calculated the primary statistic in two stages:

$$r_t = \mathrm{Pearson}(y_{t,i}, \hat{y}_{t,i})\,, \qquad \bar{r} = \frac{\sum_t n_t r_t}{\sum_t n_t}.$$

Thus, the comparison asks whether each model ranks compounds correctly *within* each kinase series and then averages the target-level statistics using the compound counts. Spearman ρ and Kendall τ provide complementary rank-based comparisons.

I estimated uncertainty with 2,000 bootstrap replicates (seed 20260817). In each replicate, I resampled compounds with replacement separately inside each target and recomputed the compound-weighted statistic.

4.2 Structure-recovery metrics on 16 complexes

I identified 16 exact FEP+4 ligand–target identities with deposited crystal structures and used the same reference and metric for FlashBind/FABind+ and Boltz-2.

For each pair, I matched the kinase chains by sequence and defined the experimental pocket as residues having any heavy atom within 5 Å of the crystal ligand. I then fitted predicted to experimental pocket using only matched protein Cα atoms. I held that transform fixed while applying it to the predicted ligand and calculated

$$\mathrm{RMSD}_{\mathrm{pose}} = \min_{\pi \in \mathcal{M}} \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left\| R\mathbf{x}_i + \mathbf{t} - \mathbf{x}^{\mathrm{ref}}_{\pi(i)} \right\|^2},$$

where R and $\mathbf{t}$ come only from the protein fit and $\mathcal{M}$ contains complete bond-order- and chirality-compatible atom mappings. The minimum over $\mathcal{M}$ corrects for chemically equivalent atom permutations. I never fitted the predicted ligand directly to the crystal ligand. I use RMSD < 2 Å only as a conventional descriptive pose threshold, not as a hypothesis-test boundary.

5 Results

Target	Pearson r	Spearman ρ	Kendall τ
CDK2	0.748	0.759	0.550
TYK2	0.917	0.921	0.767
JNK1	0.666	0.667	0.482
p38	0.869	0.894	0.723
Compound-weighted	**0.807**	**0.819**	**0.641**

Table 2: Nesso-1 released-checkpoint reproduction. The paper reports a compound-weighted Pearson correlation of 0.80 on this benchmark.

Target	Pearson r	Spearman ρ	Kendall τ
CDK2	0.831	0.865	0.667
TYK2	0.807	0.779	0.617
JNK1	0.758	0.805	0.616
p38	0.364	0.433	0.287
Compound-weighted	**0.627**	**0.666**	**0.497**

Table 3: Boltz-2 MSA-1024 released-checkpoint reproduction. The paper references are weighted Pearson $r = 0.66$ and Kendall $\tau = 0.48$; the present values are 0.627 and 0.497.

Target	Pearson r	Spearman ρ	Kendall τ
CDK2	0.612	0.641	0.433
TYK2	0.134	0.147	0.083
JNK1	0.546	0.578	0.425
p38	0.677	0.591	0.413
Compound-weighted	**0.533**	**0.515**	**0.359**

Table 4: FlashBind two-checkpoint ensemble using the authors' released FABind+ poses. The reported references are weighted Pearson $r = 0.53$ and Kendall $\tau = 0.38$; the present values are 0.533 and 0.359.

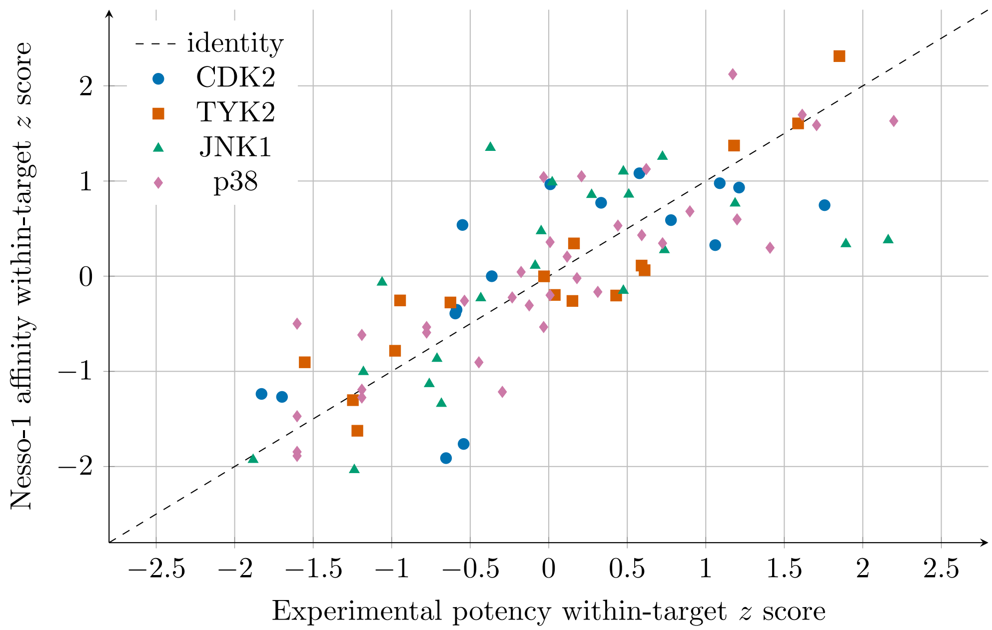

Figure 1: Within-target standardized experimental and predicted potency. Lower values correspond to stronger binding on both axes. Standardization removes target-specific offsets and scale differences for visualization only; metrics were computed from the unstandardized values.

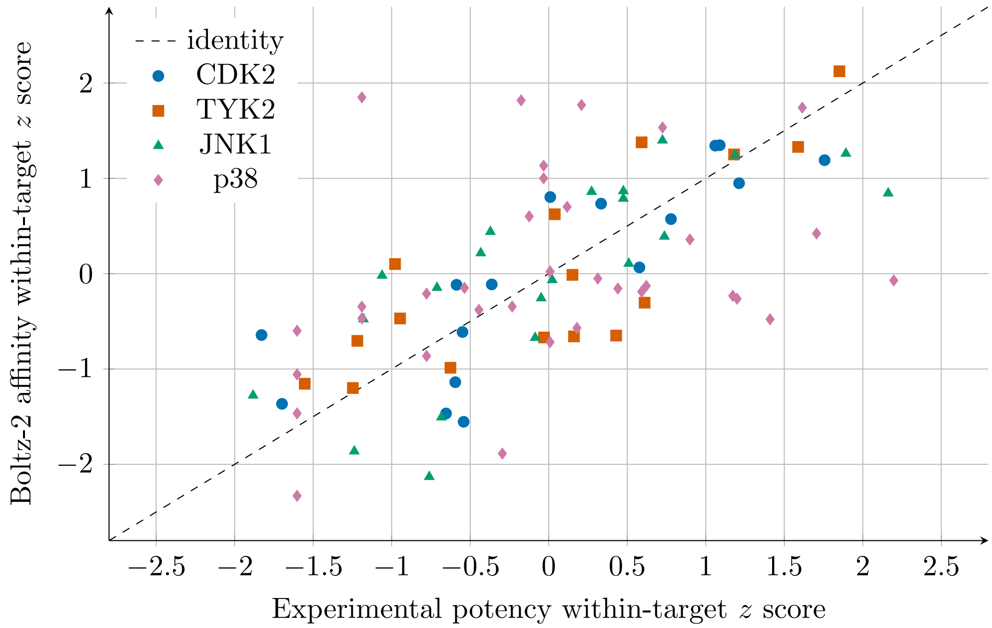

Figure 2: Boltz-2 MSA-1024 standardized scores. The dispersed p38 series is the main target-level departure from Nesso-1 and from the other Boltz targets.

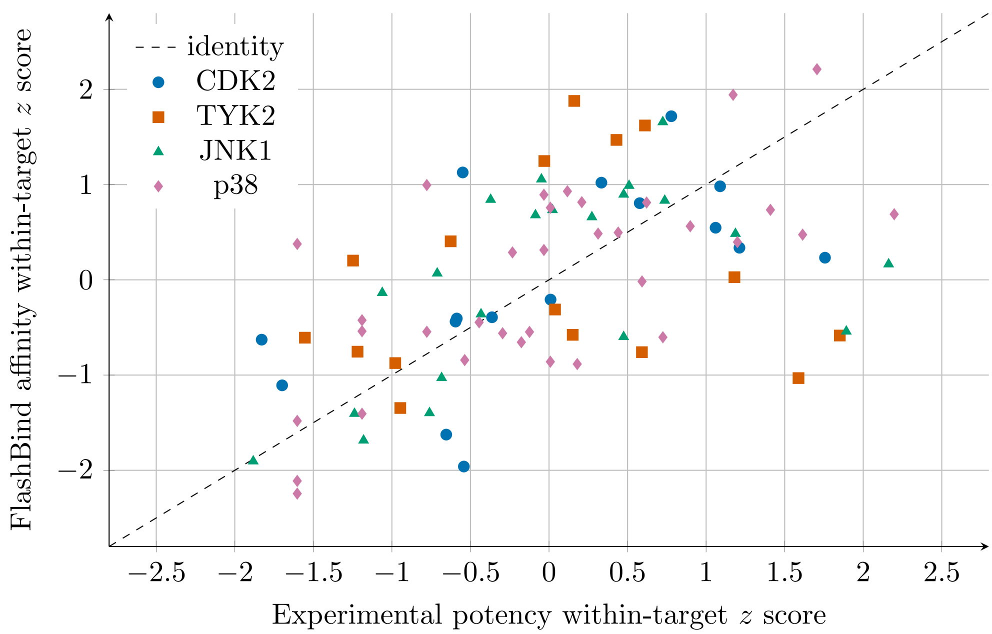

Figure 3: FlashBind standardized scores from the released FABind+ poses. TYK2 is the clear weak target, whereas the other three series retain positive within-target ranking relationships.

I obtained an aggregate Pearson result of 0.807 for Nesso (95% interval 0.731–0.873), which agrees closely with the reported 0.80. For Boltz, I obtained 0.627 (interval 0.484–0.744), which is 0.033 below the paper reference of 0.66; its Kendall result of 0.497 is 0.017 above the paper reference of 0.48. Given reconstructed preprocessing, one random seed, and an explicit MSA-depth change, I regard this as a close released-model reproduction rather than numerical identity.

For FlashBind I obtained weighted Pearson $r = 0.533$ (95% interval 0.382–0.669), essentially identical to the reported 0.53. The weighted Kendall value was 0.359, 0.021 below the reported 0.38. The close Pearson match is reassuring for software reproduction, but it is not independent validation: the authors supplied the FABind+ poses and representations used by their own scoring benchmark.

Statistic	Nesso-1	Boltz-2 MSA-1024	FlashBind
Pearson r	**0.807**	**0.627**	**0.533**
Spearman ρ	0.819	0.666	0.515
Kendall τ	0.641	0.497	0.359

Table 5: Full matched comparison across the 87-compound benchmark. Each entry is the compound-count-weighted mean of the corresponding within-target statistic.

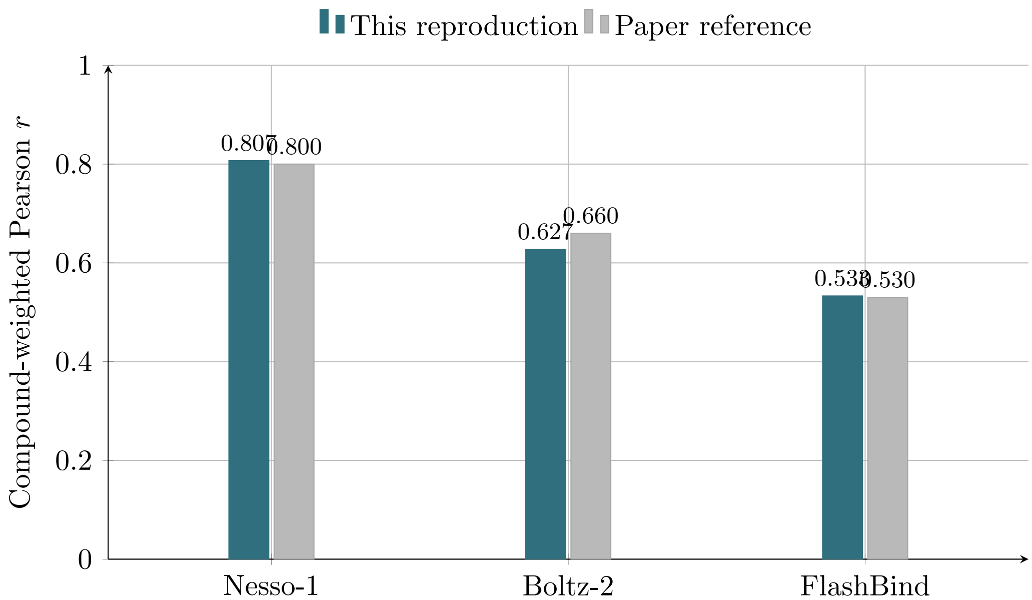

Figure 4: Weighted within-target Pearson correlations from this reproduction and the corresponding paper reference values. The reproduced values differ from the references by +0.007 for Nesso-1, −0.033 for Boltz-2, and +0.003 for FlashBind. Bars show point estimates; the bootstrap intervals are reported in the text and executive-summary table.

I found that the paired interval excludes zero for all three correlation statistics, supporting a difference in within-target ranking performance. Target-level heterogeneity is scientifically important: Boltz is better on CDK2 and JNK1, Nesso is better on TYK2, and p38 supplies the largest Nesso advantage ($r = 0.869$ versus 0.364). My experiment does not determine whether that p38 difference arises from model architecture, MSA subsampling, checkpoint stochasticity, or reconstructed input/preprocessing details. FlashBind shows a different pattern: p38 is its strongest target ($r = 0.677$), while TYK2 is much weaker ($r = 0.134$). Because this run uses released docked poses, that TYK2 result cannot be diagnosed as a fresh docking failure without separately examining the archived poses.

5.1 Protein and ligand RMSD on the 16 structure-matched pairs

RMSD component (Å)	Boltz-2		FlashBind/FABind+	
	Median	Mean	Median	Mean
Global protein Cα, fitted	1.231	1.282	0.272	0.279
Pocket protein Cα, fitted	0.400	0.998	0.208	0.220
Ligand heavy atoms after pocket fit	0.898	0.977	0.634	0.651

Table 6: Separate protein and ligand errors across the same 16 cocrystal references. No joint protein–ligand RMSD is used. The global and pocket protein rows are residual Cα RMSDs after fitting the indicated protein atoms. The ligand row is symmetry-corrected heavy-atom RMSD after applying the fixed pocket-protein transform; the ligand is not independently fitted.

For Boltz-2, the protein rows evaluate a generated receptor structure. For FlashBind, they instead measure the difference between the supplied fixed experimental receptor and each ligand-specific cocrystal receptor; they are therefore not a prediction-accuracy result for FlashBind. The separate rows avoid allowing the much larger protein atom set to dominate a combined RMSD and avoid using the experimental ligand coordinates to improve the fit.

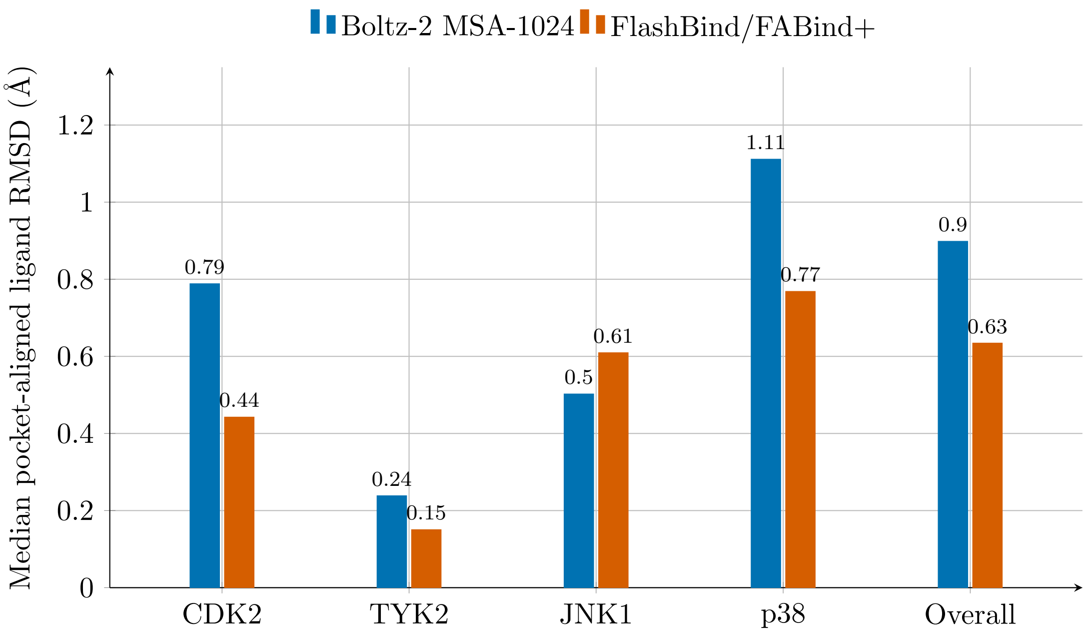

Figure 5: Median symmetry-corrected ligand RMSD by target and over all 16 structure-matched pairs. The target counts are CDK2 $n = 6$, TYK2 $n = 2$, JNK1 $n = 1$, and p38 $n = 7$; the JNK1 bar is therefore a single pair rather than a stable target-level estimate. FlashBind has the lower overall median, but the per-pair comparison remains mixed (9 lower for FlashBind and 7 for Boltz-2).

The released FlashBind/FABind+ poses were below 2 Å for all 16 pairs, whereas the Boltz-2 poses were below 2 Å for 14. FlashBind had the lower RMSD on 9 pairs and Boltz-2 on 7. The median paired FlashBind-minus-Boltz difference was -0.109 Å; its mean was -0.326 Å, influenced by the two Boltz poses above 2 Å. This pattern does not support a claim of uniform FlashBind superiority.

The protein decomposition changes the interpretation. Boltz-2's median global protein Cα RMSD was 1.231 Å, while its median locally fitted pocket Cα RMSD was 0.400 Å. The pocket result was strongly target-dependent: the Boltz medians were 0.348 Å for CDK2, 0.243 Å for TYK2, 0.460 Å for JNK1, and 2.305 Å for p38. Thus the p38 ligand-pose result coincides with a substantial local pocket-structure error and should not be interpreted as ligand placement alone. In contrast, the FlashBind receptor pocket median was 0.208 Å overall because its starting receptors were experimental structures.

I also audited the coordinate identities before scoring. Every released SDF matched its reference ligand in heavy-atom count and admitted a complete chemical graph mapping. The released p38 receptor has a formatting defect in which the chain identifier disappears after residue 168; I joined the two continuously numbered fragments in memory without changing coordinates and recorded the repair. The archive documents these ligands as FABind+ docking poses, but it does not contain the original docking commands, seeds, or per-pose logs. Thus I have evaluated the supplied poses, not independently reproduced their generation.

6 Interpretation and next-week plan

Using independently reconstructed inputs, I reproduced the Nesso-1 and Boltz-2 paper headlines to within 0.007 and 0.033 Pearson units, respectively. Using the authors' released pose archive, I reproduced FlashBind to within 0.003 Pearson units. I view these as useful but differently strong reproducibility results: FlashBind's input preprocessing is less independent. My paired comparison also supports higher average within-series ranking correlations for Nesso on these exact 87 examples, but it does not establish universal superiority or show that the model learned transferable physics. The benchmark is small, public, and congeneric; paper-level train/test chemical overlap may matter, and the Boltz run uses a documented 1,024-row trunk-MSA subsample.

The 16-pair structure analysis adds a complementary result: both methods often recover the known pose on this selected public subset, and the released FlashBind/FABind+ archive has the lower aggregate RMSD. However, FlashBind was given fixed experimental receptor structures while Boltz-2 had to generate the receptor–ligand complex. I therefore interpret the pose result as a successful archive and metric validation, not a fair end-to-end ranking of the two architectures.

Next week, I will move from benchmark reproduction to a source-code-level examination of how Boltz-2, Nesso-1, and FlashBind represent protein–ligand interactions. I will use the TYK2–4GIH complex as a common worked example and trace one forward pass through each released model, recording the residue, atom, and pair representations that connect the inputs to the affinity output.

I will then perform one small probe matched to each architecture:

- for Boltz-2, compare fixed-seed predictions as MSA information is removed or subsampled;
- for Nesso-1, compare the response to sequence changes near its predicted pocket with changes far from that pocket; and
- for FlashBind, compare experimental, released FABind+, displaced, and clashing ligand poses while also checking whole-complex rigid-body invariance.

The purpose is not to establish another performance ranking. It is to determine what information each affinity head actually receives and whether its behavior is consistent with the invariances, locality, and geometric sensitivity expected for protein–ligand binding. End-to-end FABind+ docking, additional stochastic replicates, and the p38 target-level discrepancy remain useful later experiments once this architecture audit is in place.

Reproducibility record

The tracked manifest SHA-256 is

`5f0fb23bb97ea84f3da153e050d5913b97da9bba091fb84cd556b9894fdac1e1`.

The Nesso checkpoint SHA-256 is

`9928a8a824d147d665e76656804af1cd91c86731516d0b561f9fd1c91ee45622`.

The Boltz structure and affinity checkpoint SHA-256 values are

> Structure: `090e82ac8c92f5e943fa1b39e7410a44027bea7243c0bbb3caa67a77fc1428e1`
> Affinity: `dcc5cd3722b1c9eaa34267e4ae32f55cbbf1963f4c19319381ccfa30fdd2ca9e`.

The two FlashBind value-checkpoint SHA-256 values are

> Value 1: `ca87f84dedd4642da693a508b1119c802caa370e356dad31c47b8a708821e553`
> Value 2: `59fa2686b0859a632862aec21ea69f2bebb1ab104e48606fe739c929368d488d`.

All commands, environments, timestamps, hardware samples, return codes, raw outputs, and parsing status are stored beneath the ignored experiment run tree. Compact metrics and this report are tracked in Git.

References

[1] Songlin Jiang, Yifan Chen, Aarti Krishnan, Yu Zhang, and Wengong Jin. Flashbind: Towards accurate and efficient structure-based virtual screening. *bioRxiv*, 2026. doi: 10.64898/2025.12.22.695983. URL https://doi.org/10.64898/2025.12.22.695983. Preprint, version 2, posted 8 April 2026; version 1 was titled FlashAffinity.

[2] Nikhil Shenoy, David Errington, Emmanuel Bengio, Kacper Kapuśniak, Kerstin Klaeser, Yui Tik Pang, Vladimir Radenkovic, Prudencio Tossou, Therence Bois, Andrew Wedlake, and Francesco Di Giovanni. Nesso-1: Accelerating open-source binding affinity predictions. *bioRxiv*, 2026. doi: 10.64898/2026.08.01.742196. URL https://doi.org/10.64898/2026.08.01.742196. Preprint, version 1, posted 3 August 2026.

[3] Saro Passaro, Gabriele Corso, Jeremy Wohlwend, Mateo Reveiz, Stephan Thaler, Vignesh Ram Somnath, Noah Getz, Tally Portnoi, Julien Roy, Hannes Stark, David Kwabi-Addo, Dominique Beaini, Tommi Jaakkola, and Regina Barzilay. Boltz-2: Towards accurate and efficient binding affinity prediction. *bioRxiv*, 2025. doi: 10.1101/2025.06.14.659707. URL https://doi.org/10.1101/2025.06.14.659707. Preprint, version 1, posted 18 June 2025.

[4] David F. Hahn et al. Best practices for constructing, preparing, and evaluating protein-ligand binding affinity benchmarks. *Living Journal of Computational Molecular Science*, 4(1):1497, 2022. doi: 10.33011/livecoms.4.1.1497. URL https://doi.org/10.33011/livecoms.4.1.1497.